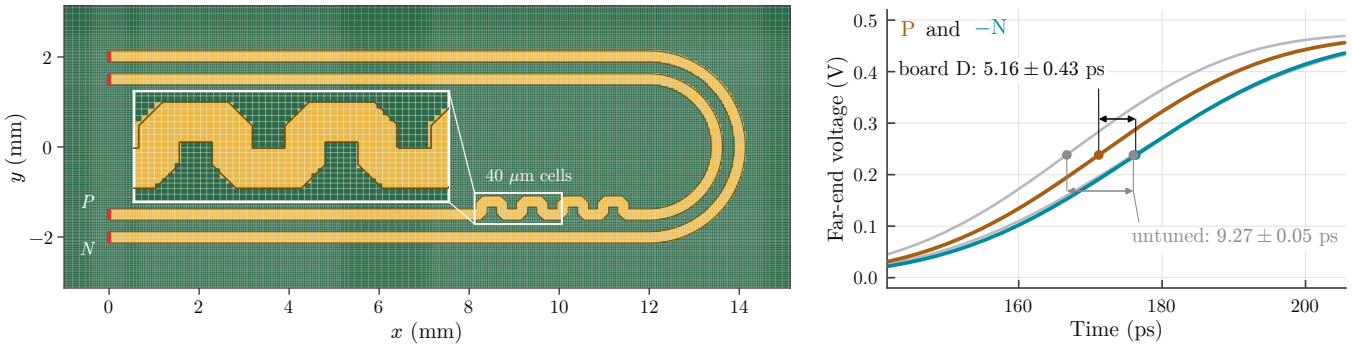
Beyond Length Matching: Electromagnetic Verification and Optimization of Differential-Pair Meanders

Animesh Chouhan
Copperhead
animesh@copperhead.sh

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(a) (b)

Figure 1: Board D, one of the study's fixtures. (a) Seen from above, to its outline, as openEMS meshes it: the substrate (green), the copper (gold) and the four 50 Ω lumped ports (red), with the medium mesh the runs used drawn over it, graded from 40 μm cells over the copper to 150 μm away from it. Cells whose centre lies in the copper are filled; the inset enlarges two of the four excursions so the staircase shows. The excursions make P exactly as long as N. (b) What openEMS finds at the receiver for a 50 ps edge: P still arrives 5.16 ± 0.43 ps before N, against 9.27 ps for the untuned bend (grey).

Abstract

A 180-degree U-bend in a differential pair makes the inner trace shorter by πs, and the usual fix is a serpentine that adds the same length back. We test that fix with full-wave simulation instead of assuming it. A controlled four-port fixture ($s = 0.508$ mm, 20 mil) is solved in openEMS (FDTD) for 22 geometries in microstrip and stripline, 140 single-port-excited runs in all, and every result is taken through mixed-mode S-parameters to receiver waveforms. The untuned bend gives 9.27 ps of skew, against 9.45 ps from geometry and the single-line effective permittivity. Tuning that makes the two centrelines exactly equal in length removes only 44–75 % of that skew. 2.29–5.16 ps remain, with a numerical uncertainty of ±0.31 to ±0.43 ps, and the shortfall grows with the number of excursions. A transmission-line model built from the same lengths predicts less than 1 ps. A serpentine experiment attributes the gap to the corners of each excursion and to coupling between its folds (delay efficiency 0.84 with folds 1.0 mm apart, 0.68 at 0.15 mm). The same copper in stripline behaves the same way, so the shortfall is not a microstrip effect. At 16 Gb/s the skew hardly closes the differential eye, from 92.9 to 91.1 % of the swing, but turns 13.6 % of the swing into common mode. Sizing the tuning from its own residual in one more EM iteration brings the skew to 0.15 and 0.56 ps for the broad and sparse tunings, but only to 1.90 ps for the dense accordion, whose added amplitude couples across its folds too. All of this is simulation. A coupon plan for measuring it is given.

1 Introduction

Length matching is the standard answer to intra-pair skew. A widely shared illustration [1] shows the inner trace of a U-bend arriving about 9.5 ps early at 20 mil pitch, corrected by a serpentine of about 1.6 mm. The geometry is right, and the delay estimate is reasonable for a common stackup. What the picture does not establish is the step from equal *length* to equal *delay*. A serpentine pulls one trace away from its partner, which changes the coupling, and it adds corners and folds that couple to each other. A CAD length report shows none of that.

This paper asks one question. When a bent pair is tuned to equal centreline length, how much of the skew is actually removed, and at what cost in reflection, mode conversion and board area? We answer it for one microstrip stackup by full-wave simulation of a controlled fixture and check it in stripline. A transmission-line model sits beside the full-wave results to separate what length explains from what it does not (Fig. 8).

Contributions: (i) a fixture and parameter set in which

only the tuning changes (Section 3); (ii) four-port full-wave extraction with mesh convergence and an uncertainty in picoseconds (Sections 4, 6); (iii) the simulated delay efficiency of four tuning styles and two serpentines, in both media (Section 7); (iv) an EM-informed sizing step and a Pareto comparison (Section 8); (v) a coupon plan to check the predictions in hardware (Section 9). The scripts and data are released with the report (Section 11).

What is and is not claimed. Every curve and number in Sections 7 and 8 is a simulation result. Section 3 is analytical and is labelled so. No hardware was built and nothing was measured. Mode conversion is reported as S_{cd21} and as common-mode voltage at the receiver. Neither is radiated emission, and the paper makes no emission claim.

2 Background

A symmetric coupled pair carries two quasi-TEM modes. The odd mode carries the differential signal and the even mode the common mode. In microstrip the odd mode has more of its field in air, so its effective permittivity is lower and it travels faster [2]. A length mismatch between the two traces converts part of a differential wave into common mode, which then travels at the even-mode velocity.

Mixed-mode S-parameters [3, 4] express this directly. S_{dd21} is differential transmission, S_{dd11} differential reflection, and S_{cd21} the common mode that leaves port 2 when differential mode enters port 1. For a pair whose two paths differ only by a delay $\Delta\tau$ and whose modes do not interact,

$$\left|\frac{S_{cd21}}{S_{dd21}}\right| = \left|\tan\frac{\omega\Delta\tau}{2}\right|, \tag{1}$$

which we invert to give a frequency-resolved skew. Earlier work on single-ended serpentines found that coupling between adjacent segments makes a meander electrically shorter than its length [5, 6]. We test whether that matters at the scale of a single bend correction, and in a pair.

3 Geometry and analytical estimates

The bend. Two concentric arcs that turn through θ with centreline spacing s differ in length by

$$\Delta L = (R+s)\,\theta - R\,\theta = s\,\theta, \tag{2}$$

independent of the radius R (Fig. 2). For a single 180-degree U-turn and $s = 20\,\text{mil} = 0.508\,\text{mm}$, $\Delta L = \pi s = 1.596\,\text{mm} = 62.8\,\text{mil}$. Two 180-degree turns in the same direction double this ($2\pi s$). Two in opposite directions cancel in length, but the pair carries the skew, and its

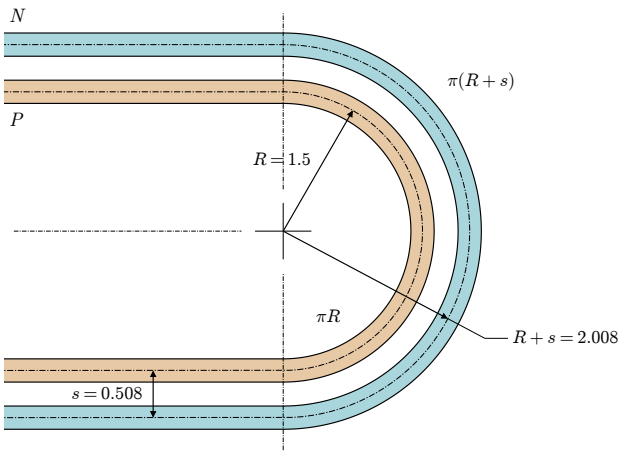


Figure 2: The 180-degree bend of fixture B, dimensions in mm. Centrelines dash-dotted; P is the inner trace. The arc lengths πR and $\pi(R+s)$ differ by πs whatever the radius.

common mode, between them. The nominal delay difference is

$$\Delta t = \Delta L\sqrt{\varepsilon_{\text{eff}}}/c. \tag{3}$$

With $\varepsilon_{\text{eff}} = 3.8$, Eq. (3) gives 10.4 ps, and the illustrated 9.5 ps corresponds to $\varepsilon_{\text{eff}} = 3.18$, a faster line. For our stackup the closed form of Hammerstad and Jensen [7] gives a single (uncoupled) line $\varepsilon_{\text{eff}} = 3.154$ and $Z_0 = 64.2\,\Omega$, so $\Delta t = 9.45$ ps. The odd- and even-mode values we extract below ($\varepsilon_{\text{eff}} = 2.97$ and 3.50) bracket this at 9.17 and 9.95 ps. Eq. (3) uses one velocity, and a coupled pair has two. That is the first thing it omits.

The tuning rule. N rectangular excursions of amplitude A (Fig. 3) add

$$\Delta L_{\text{add}} \approx 2NA \tag{4}$$

of centreline. Eq. (4) omits five things. The first is geometry, the length each corner treatment removes (a 45-degree chamfer of leg c removes $(2-\sqrt{2})c$ per corner, $4(2-\sqrt{2})c$ per excursion), and we correct for it. The other four are electrical: the change of mode where the tuned trace leaves its partner, the change in differential impedance over that stretch, the corner as a discontinuity whose electrical length differs from its centreline, and coupling between the folds of the excursion. In our fixtures the amplitude is solved so that the two centrelines are equal to well under a micrometre after the chamfers are cut. The 2NA rule alone would leave the dense accordion 0.70 mm short.

The fixture. One microstrip layer over a return plane (Fig. 5, Table 1): $h = 0.2$ mm, $\varepsilon_r = 4.3$, $w = 0.25$ mm, $s = 0.508$ mm. Every bent variant uses the same U-bend ($R = 1.5$ mm for P), the same 12 mm legs and the same launch plane, so only the tuning differs (Fig. 4): A, a straight pair of the U-bend's mean length; B, the untuned bend; C, two sparse excursions next to the bend; C far, the same two excursions next to the launch; D, a dense accordion of four; F, one broad trombone; and E, a pair-level raised-cosine excursion of both traces together,

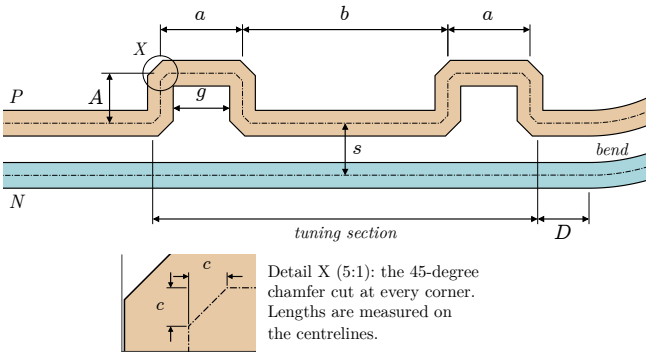


Figure 3: One tuning section (C, $N = 2$) and the symbols used for it: amplitude A between centrelines, top a and gap b along them, fold edge spacing g, pitch s, distance to the bend D. Every corner carries a 45-degree chamfer of leg c (detail X, 5:1).

Table 1: Parameters. All values are **proposed** for this study; none is calibrated to a laminate or a fabricator.

Parameter	Unit	Value	Range or note
Dielectric height h	mm	0.2	fixed
Relative permittivity ε_r	–	4.3	fixed
Loss tangent at 5 GHz	–	0.02	fixed, $\propto 1/f$
Copper thickness / σ	mm / MS/m	0.035 / 58	sheet model
Trace width w	mm	0.25	fixed
Pair pitch s (centre to centre)	mm	0.508	fixed, 20 mil
Bend radius of P, R	mm	1.5	0.75, 1.5, 3.0
Leg length (launch to bend)	mm	12.0	fixed
Excursions N	–	1, 2, 4	F, C, D
Amplitude A	mm	solved	0.29–1.08
Excursion top / gap	mm	0.45–3.0	per family
Corner chamfer	mm	0.075	fixed
Stripline plane spacing	mm	0.7	trace midway
Distance to bend D	mm	0.5	0.5 or at launch
Serpentine added length	mm	6.0	fold gap 1.0, 0.15
Edge (10–90 %)	ps	35–300	post-processing
Bandwidth	GHz	0–30	Gaussian pulse
Finest cell (coarse/medium/fine)	µm	60/40/27	convergence

a control whose net turning angle is zero and which therefore leaves the length difference unchanged. Each single-trace tuning is cut into P on the leg leading into the bend and turns away from N. Every route length change is intentional and listed in Table 2. Clearance between P and N copper is checked by our generator and by KiCad 10's DRC (`kicad-cli pcb drc`, 0.15 mm rule), and no variant has a clearance or short violation. KiCad does not check spacing within one net, so the fold spacing is set by the parameters, 0.20 mm edge to edge in D.

4 Numerical model

4.1 Stage II: a two-conductor line model

For the pair, the coupled telegrapher equations are

$$\frac{\mathrm{d}\mathbf{V}}{\mathrm{d}x} = -(\mathbf{R} + j\omega\mathbf{L})\,\mathbf{I}, \qquad \frac{\mathrm{d}\mathbf{I}}{\mathrm{d}x} = -(\mathbf{G} + j\omega\mathbf{C})\,\mathbf{V}, \quad (5)$$

with 2×2 per-unit-length matrices whose off-diagonal terms are the coupling [2]. For a symmetric pair the even and odd modes diagonalise them. Rather than assume a cross-section, we calibrate the modes on the full-wave straight fixture A. Its S_{dd} (100 Ω) and S_{cc} (25 Ω) are each the two-port of a uniform line, and the extraction of Eisenstadt and Eo [8] gives $Z_{\mathrm{odd}} = 53.4\,\Omega$ ($Z_{\mathrm{diff}} = 106.7\,\Omega$), $Z_{\mathrm{even}} = 67.7\,\Omega$ ($Z_{\mathrm{comm}} = 33.8\,\Omega$), $\varepsilon_{\mathrm{eff}} = 2.97$ (odd) and 3.50 (even), with loss and dispersion over the whole band (the values quoted are means from 0.5 to 10 GHz). Each variant is then a cascade of chain matrices, coupled where the traces run together and uncoupled single lines (Hammerstad–Jensen parameters) where P leaves N or where N's excess lies in the bend, spread over 18 slices. The model carries every length and none of the rest. Corners, coupling between a meander's folds or across the U, and the bend as a discontinuity are all missing from it, so where it disagrees with full wave the cause is something other than length.

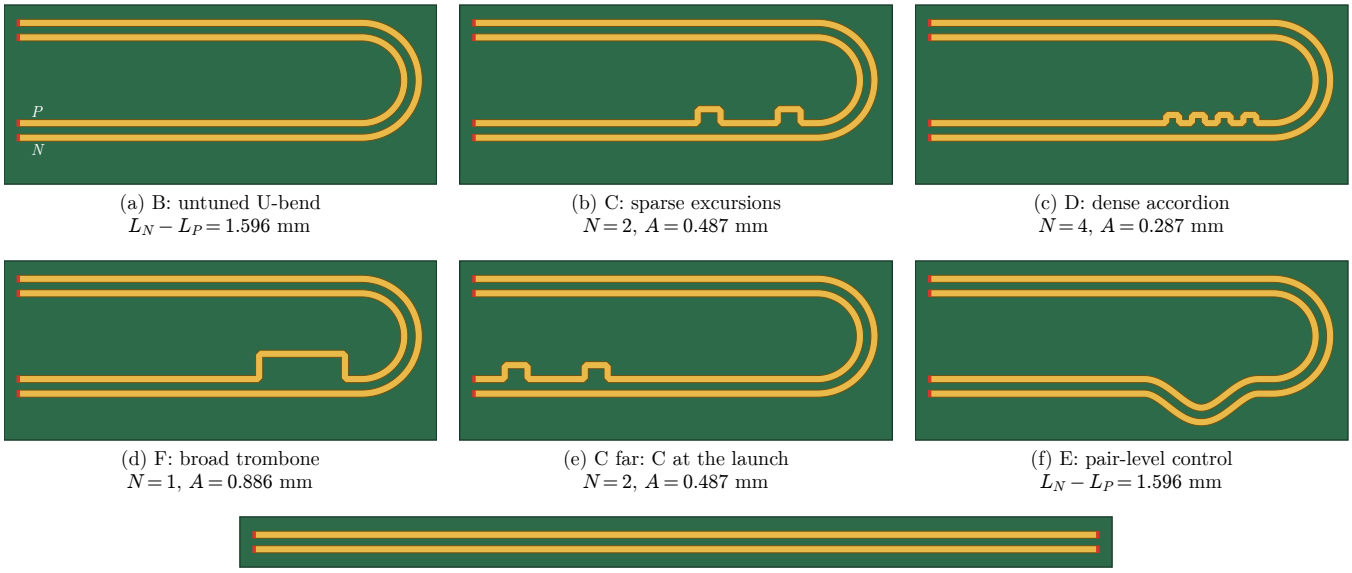
4.2 Stage III: full-wave extraction

The solver is openEMS [9, 10], an equivalent-circuit FDTD code, version `v0.37.0-rc3-31-g6761a36`. Its Python bindings were not installed on the host, so the model is written directly in the XML vocabulary those bindings produce, which we had checked against this openEMS version before. The signal copper is a conducting sheet (35 µm, 58 MS/m) drawn from the generator's polygons, the return plane is the PEC floor of the domain, and the dielectric is a lossy box (Table 1). openEMS models the loss as a fixed conductivity, so $\tan\delta$ falls as $1/f$. The other five faces are first-order Mur absorbing boundaries [11], at least 2 mm ($10h$) from any copper laterally and 2.5 mm above it (Fig. 7).

Each trace end has a 50 Ω lumped port from the plane to the copper, on the launch plane $x = 0$ (Fig. 6). Ports are numbered in scikit-rf's mixed-mode order [12]: 1 = P near, 2 = N near, 3 = P far, 4 = N far. Each variant takes four runs, each exciting one port with the other three passive [13], with a Gaussian pulse covering 0 to 30 GHz. From each port's voltage u_k and current i_k (into the structure), the power waves [14] $a_k = (u_k + Z_0 i_k)/(2\sqrt{Z_0})$ and $b_k = (u_k - Z_0 i_k)/(2\sqrt{Z_0})$ give $S_{kj} = b_k/a_j$. The result is written as a Touchstone `.s4p` from DC to 30 GHz in 10 MHz steps.

The mesh is rectilinear and graded, with the finest cells over the copper regions (60, 40 and 27 µm for the coarse, medium and fine levels, with every straight trace edge on a grid line), growing by at most 1.3 per cell to 0.15 mm laterally and 0.3 mm in the air, limits that are the same at every level, so refinement is of the copper regions only. All bent variants share one mesh per level, built over the union of their copper, so discretisation error in the copper they share is common to them. The medium mesh of the bent fixtures has 1.24 M cells and the fine one 2.96 M. The arcs and chamfers are staircased (Fig. 6b).

The pulse's DC content leaves a static field in the domain whose energy settles near −38 dB, so no run reaches an energy end criterion. A pulse without DC content does reach one, but it measures nothing useful below about 2 GHz. Every run therefore stops at a fixed 2 ns of simulated time, by which the port voltages have fallen at least 98 dB below their peak (Table 3). On fixture B



(a) B: untuned U-bend
$L_N - L_P = 1.596$ mm

(b) C: sparse excursions
$N = 2$, $A = 0.487$ mm

(c) D: dense accordion
$N = 4$, $A = 0.287$ mm

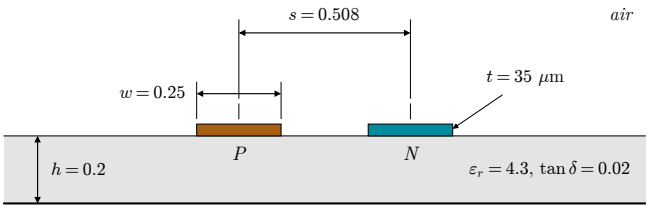
(d) F: broad trombone
$N = 1$, $A = 0.886$ mm

(e) C far: C at the launch
$N = 2$, $A = 0.487$ mm

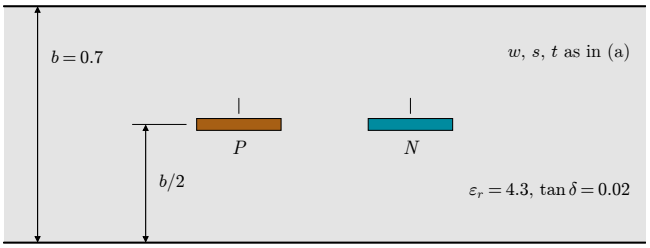
(f) E: pair-level control
$L_N - L_P = 1.596$ mm

(g) A: straight reference, the U-bend's mean length

Figure 4: The variants as openEMS is given them, all at one scale: the copper polygons (gold) and the four 50 Ω lumped ports (red) on the substrate. In every tuned variant (b–e) P and N are exactly equal in centreline length; N is the number of excursions and A the solved amplitude. A (g) is straight, as long as the U-bend's mean centreline.



(a) Microstrip



(b) Stripline

Figure 5: Cross-sections at true scale, dimensions in mm. The stripline variants are the same copper midway between two planes.

(coarse mesh, port 1 excited) a 7 ns record changes the low-frequency S-parameters by about 10^{-3}.

5 Analysis metrics

Mixed mode. `se2gmm(p=2)` on the order above gives $d_{\text{near}}, d_{\text{far}}, c_{\text{near}}, c_{\text{far}}$, so $S_{dd11} = S_{00}$, $S_{dd21} = S_{10}$, $S_{cd21} = S_{30}$ (zero-based), with reference impedances of 100 Ω differential and 25 Ω common, in scikit-rf [15]. We checked this convention against the scikit-rf 2.2.0 source rather than its rendered documentation.

Waveforms. A differential step of 1 V open circuit (+1 V and −1 V, each behind 50 Ω) with a Gaussian edge of 10–90 % rise time t_r drives ports 1 and 2, and ports 3 and 4 are 50 Ω to the plane (100 Ω differential, 25 Ω common). Far-end voltages follow by superposition from the four-port in the frequency domain and an inverse FFT, with nothing fitted. $V_d = V_P - V_N$, $V_c = (V_P + V_N)/2$.

Skew is $t_{50}(N \text{ falling}) - t_{50}(P \text{ rising})$ at the far end, each at half its own final value. Positive means P, the inner trace, arrives first. We report 50 ps edges unless stated, and 35 to 300 ps in Section 7. Edges faster than 35 ps are not evaluated, because the simulated band ends at 30 GHz.

Delay efficiency.

$$\eta = \frac{t_{\text{skew,ref}} - t_{\text{skew}}}{\Delta L_{\text{add}} \sqrt{\varepsilon_{\text{eff}}^{\text{single}}}/c}, \tag{6}$$

where the reference is the identical untuned fixture (B, or the straight serpentine reference), ΔL_{add} is the centreline the tuning adds to P, and $\varepsilon_{\text{eff}}^{\text{single}} = 3.154$ (5.924 ps/mm) is the velocity a designer's length-to-delay conversion would use. $\eta = 1$ means the delay is what the length predicts. On the same basis the untuned bend has $\eta = 0.981$.

Other metrics, each over 0.1–20 GHz unless stated: worst $|S_{dd11}|$; RMS $|S_{cd21}|$; differential TDR impedance from the step response of S_{dd11} with a 35 ps edge; the frequency-resolved skew of Eq. (1); a PRBS7 eye at 10 Gb/s (50 ps edges) and 16 Gb/s (35 ps edges), with height at the best phase and width where the eye is open by more than 10 % of the swing; and the board area of the tuning, its length times its amplitude. No interface specification is assumed, so wherever a pass or fail would need one the target is illustrative and is to be selected

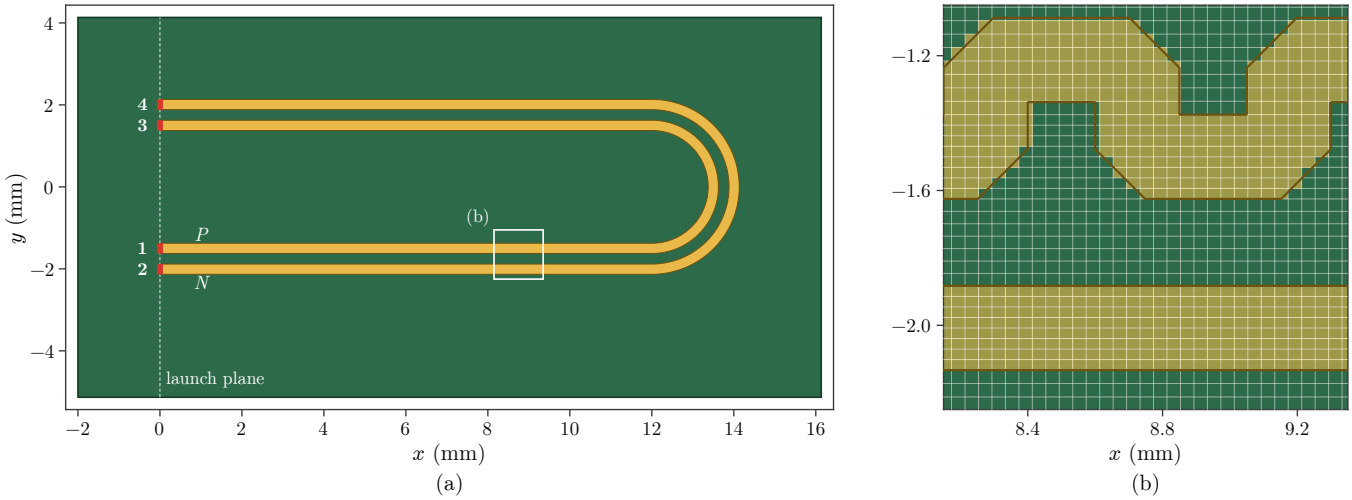


Figure 6: (a) Fixture B in its computational domain: the substrate extends 2 mm past the copper on every side, and the ports, numbered in mixed-mode order, sit on the launch plane. (b) The medium mesh (40 μm finest cell) over D's first excursions. Cells whose centre lies in the copper are filled, so the 45-degree chamfers are staircased.

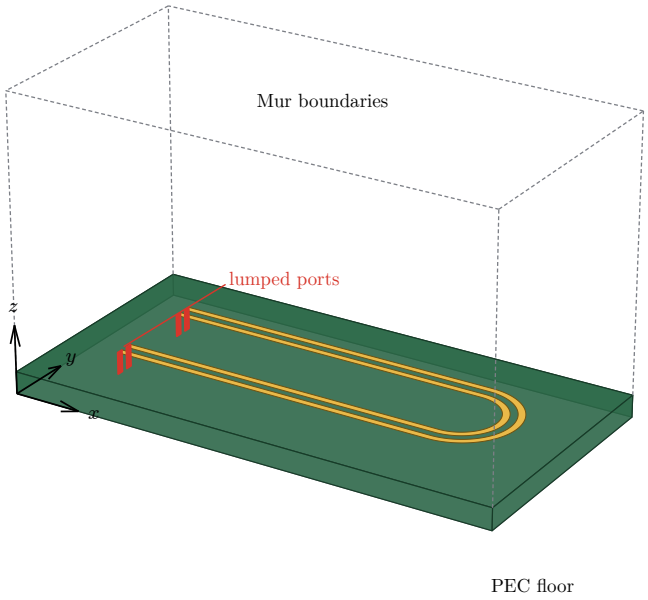


Figure 7: The model of fixture B in three dimensions, height drawn 4 times its true scale: the dielectric slab on the PEC floor, the copper sheets, the lumped ports from the floor to the copper, and the edges of the air box, whose five open faces are Mur boundaries.

for the intended interface.

6 Verification of the numerics

Table 3 lists the checks for every variant. Reciprocity holds to $|S_{ij} - S_{ji}| \leq 4.5 \times 10^{-3}$ (the straight fixture, symmetric by construction, to 3.3×10^{-7}). Every four-port is passive, with a largest singular value of S over the band of 0.989 for B and no more than 0.990 for any variant. Every record ends with its port voltages at least 98 dB below their peak.

Convergence. B, C, D and the serpentine set were solved on three meshes, and D on a fourth (Table 4, Fig. 10). We report Roache's grid convergence index [16] over the three finest meshes, with safety factor 1.25 and an observed order capped at the scheme's formal order of 2. Where those meshes do not converge monotonically we use three times their spread. B converges monotonically, with a formal GCI of 0.02 ps on the finest mesh. D's 75 μm chamfers span barely one cell on the coarse mesh, and its coarse value is the outlier. Over the three finest meshes it converges monotonically, to 4.82 ps. The tight serpentine's 0.15 mm folds still oscillate across meshes. Two caveats apply to every bound. The outer cell limits are the same at every level, so only the copper regions are refined. And where the meshes do converge monotonically the observed order is 6.0 for B and 5.3 for D, well above the scheme's formal order of 2. The three levels are therefore not in the asymptotic range, which is why the order is capped, and the finest-mesh GCIs are formal values rather than demonstrated ones. One cause is that only the straight legs have their copper edges on mesh lines. The tops and verticals of the tuning sections do not, so each level sees an amplitude and a vertical's width quantized to its own cell, by up to half a cell. The comparison tables quote the medium mesh throughout, so each medium value carries its own bound, 1.25 times its distance from the extrapolated value, or three times the spread where the meshes oscillate (Table 4). That gives ± 0.05 ps for B, ± 0.31 ps for C and ± 0.43 ps for D, whose medium value runs high by about that much. A variant solved on the medium mesh only carries the bound of the converged variant whose smallest features it shares: C's for C far, F and the EM-sized C and F; D's for the EM-sized D; B's for E and the other bend radii. Because a wide bound can hide a trend or invent one, we also judge every conclusion mesh by mesh, and Table 5 gives η on each mesh, each against its untuned reference on the same mesh.

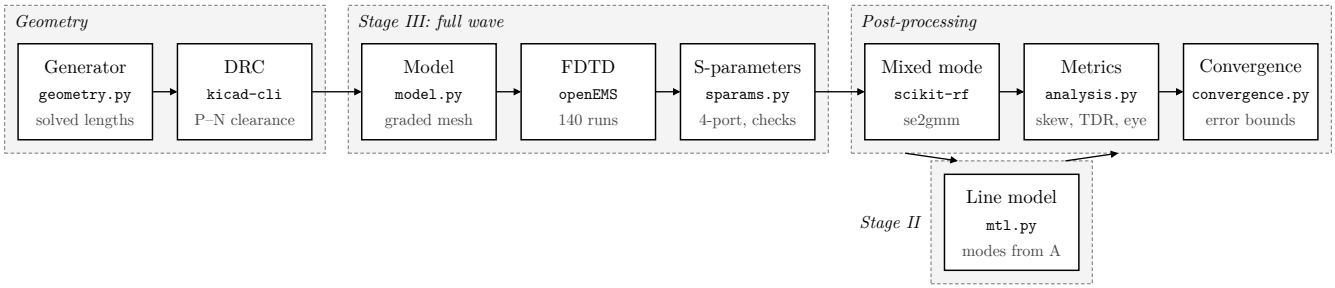


Figure 8: The pipeline. The line model (Stage II) is calibrated on fixture A's mixed-mode parameters and read beside the full-wave results. Each run keeps its model, its solver log and a provenance record (solver version, model hash, wall time, timesteps); each variant keeps its `.s4p` and its checks.

Table 2: Metrics, medium mesh, 50 ps edge. ΔL_{add} is the centreline the tuning adds to P; η is Eq. (6); MTL is the line model's skew; $|S_{dd11}|$ and $|S_{cd21}|$ over 0.1–20 GHz; Z_{diff} from TDR. Skew bounds: each variant's own GCI where it was solved on three meshes, else that of the converged variant whose smallest features it shares (Section 6).

Variant	ΔL_{add} (mm)	Skew (ps)	η	MTL (ps)	V_c p-p (%)	$\lvert S_{dd11}\rvert$ max (dB)	$\lvert S_{cd21}\rvert$ RMS (dB)	Z_{diff} max (Ω)	Area (mm^2)
A straight	–	0.00 ± 0.05	–	0.00	0.00	-23.4	–	106.0	0.00
B untuned	–	9.27 ± 0.05	–	8.83	4.87	-20.7	-10.2	107.1	0.00
E pair exc.	–	9.39 ± 0.05	–	8.83	4.92	-21.4	-10.1	108.1	4.00
C sparse	1.596	3.59 ± 0.31	0.60	0.31	1.83	-21.6	-18.0	106.9	1.83
C far	1.596	4.36 ± 0.31	0.52	0.96	2.19	-19.6	-15.4	106.9	1.83
D dense	1.596	5.16 ± 0.43	0.44	0.27	2.67	-19.8	-14.9	106.9	0.95
F trombone	1.596	2.29 ± 0.31	0.74	0.23	1.16	-16.8	-22.0	114.4	2.79
C, EM-sized	2.213	0.56 ± 0.31	0.66	-2.98	0.52	-21.7	-25.2	106.7	2.40
D, EM-sized	2.484	1.90 ± 0.43	0.50	-4.50	0.98	-19.9	-22.6	106.7	1.31
F, EM-sized	1.991	0.15 ± 0.31	0.77	-1.89	0.40	-16.7	-27.8	115.0	3.41

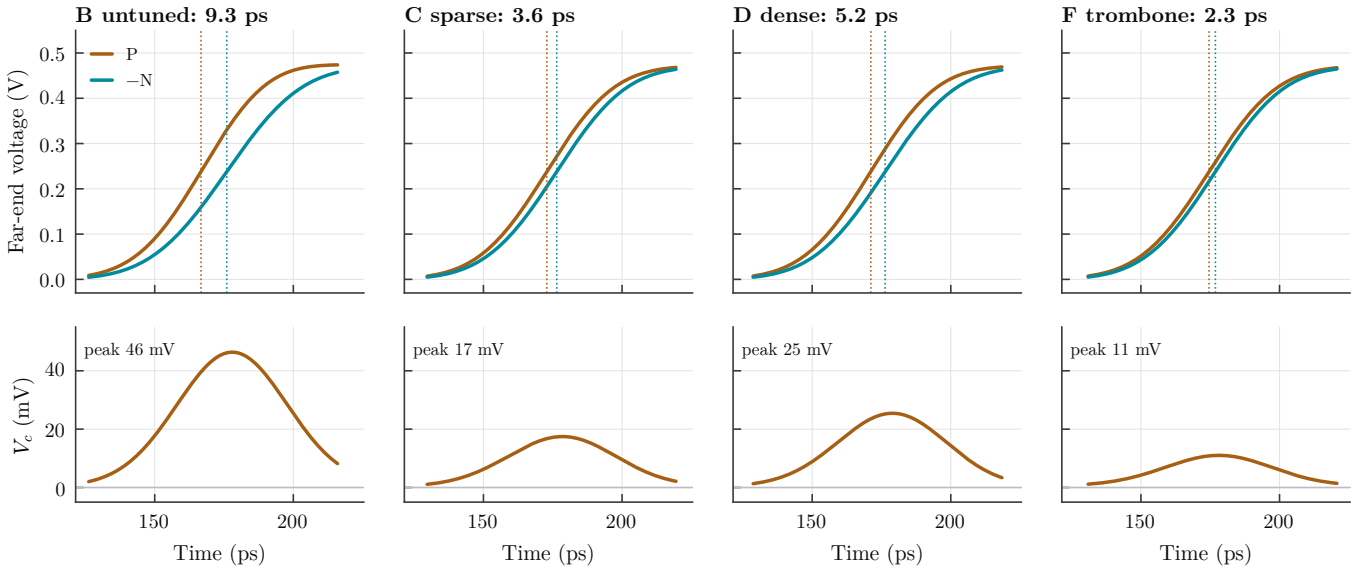


Figure 9: Far-end P and −N for a differential step (50 ps edge), and the common-mode voltage. Dotted lines mark each 50 % crossing.

7 Results

All values in this section are simulated. Table 2 collects them.

7.1 The untuned bend

B shows 9.27 ps of skew at 50 ps edges, against 9.45 ps from Eq. (3) with the single-line ε_{eff}, so geometry predicts the bend to 1.9 % by this metric. The line model, which holds the same lengths and the two modal velocities and nothing else, reads 8.83 ps by the same metric. Part of what a 50 % crossing makes of a pure length difference in a coupled pair is the mode conversion that difference causes, so the agreement with Eq. (3) is specific to the metric. The skew does depend on the edge, 8.92 ps at 35 ps and 10.03 ps at 300 ps (Fig. 19), moving from

Table 3: Numerical checks (medium mesh). Truncation is how far every port voltage had fallen, relative to its peak, over the last 5 % of the record.

Variant	Reciprocity $\max \lvert S_{ij} - S_{ji} \rvert$	Passivity $\max \sigma(S)$	Truncation (dB)
A straight	3.3×10^{-7}	0.989	-110
B untuned	2.4×10^{-3}	0.989	-110
C sparse	4.5×10^{-3}	0.988	-110
C far	2.8×10^{-3}	0.988	-109
D dense	3.4×10^{-3}	0.988	-110
E pair exc.	2.9×10^{-3}	0.988	-110
F trombone	2.8×10^{-3}	0.988	-110
C, EM-sized	3.7×10^{-3}	0.988	-110
D, EM-sized	3.0×10^{-3}	0.988	-110
F, EM-sized	2.0×10^{-3}	0.988	-110
Serp. ref.	6.0×10^{-5}	0.989	-110
Serp. 1.0 mm	1.8×10^{-3}	0.988	-109
Serp. 0.15 mm	2.1×10^{-3}	0.988	-109
B, R 0.75	3.5×10^{-3}	0.990	-110
B, R 3.0	2.4×10^{-3}	0.987	-108
B stripline	1.8×10^{-3}	0.989	-109
C stripline	1.4×10^{-3}	0.988	-109
D stripline	1.8×10^{-3}	0.988	-110
F stripline	2.2×10^{-3}	0.988	-109
Serp. ref. SL	2.9×10^{-5}	0.988	-99
Serp. 1.0 mm SL	1.5×10^{-3}	0.987	-98
Serp. 0.15 mm SL	1.5×10^{-3}	0.987	-98

Table 4: Mesh convergence of the skew (50 ps edge). The finest-mesh GCI is the formal value; the 40 µm bound is what the comparison tables carry (Section 6).

Variant	60 µm (ps)	40 µm (ps)	27 µm (ps)	18 µm (ps)	GCI finest	40 µm bound	Convergence
B untuned	9.54	9.27	9.25	–	±0.02	±0.05	monotonic
C sparse	3.69	3.59	3.61	–	±0.31	±0.31	oscillatory
D dense	4.45	5.16	4.88	4.84	±0.03	±0.43	monotonic
Serp. ref.	0.03	0.03	0.03	–	±0.00	±0.00	monotonic
Serp. 1.0 mm	-32.41	-29.99	-29.69	–	±0.31	±0.68	monotonic
Serp. 0.15 mm	-23.63	-24.12	-22.98	–	±3.40	±3.40	oscillatory

Table 5: Delay efficiency η (Eq. 6) on each mesh, against the untuned reference on the same mesh; D at 18 µm, which has no B at that level, is against B at 27 µm.

η on each mesh	60 µm	40 µm	27 µm	18 µm
C sparse	0.62	0.60	0.60	–
D dense	0.54	0.44	0.46	0.47
Serp. 1.0 mm	0.91	0.84	0.84	–
Serp. 0.15 mm	0.67	0.68	0.65	–

near the odd-mode prediction (9.17 ps) toward the even-mode one (9.95 ps) as the edge slows. The bend radius matters a little where Eq. (2) says it should not, with 9.60 ps at $R = 0.75$ mm, 9.27 ps at 1.5 mm and 9.07 ps at 3 mm (Fig. 20). The pair-level excursion E leaves the skew where B has it (9.39 ps) and adds 4.3 ps of pair delay, as designed.

7.2 Equal length is not equal delay

Fig. 11 is the central result. Every single-trace tuning makes P and N equal in centreline length, and every one leaves skew, 2.29 ps for the broad trombone F, 3.59 ps for the two sparse excursions C (±0.31 on each), and 5.16 ps for the dense accordion D (±0.43; 4.82 ps converged). In efficiency terms (Eq. (6)), $\eta = 0.74$, 0.60 and 0.44 for one, two and four excursions. The transmission-line model, given the same lengths, predicts 0.23, 0.31 and 0.27 ps. The waveforms (Fig. 9) show the same thing at the receiver. P still leads N, and the common-mode pulse falls from 4.87 % of the step for B to 1.16–2.67 %, not to zero.

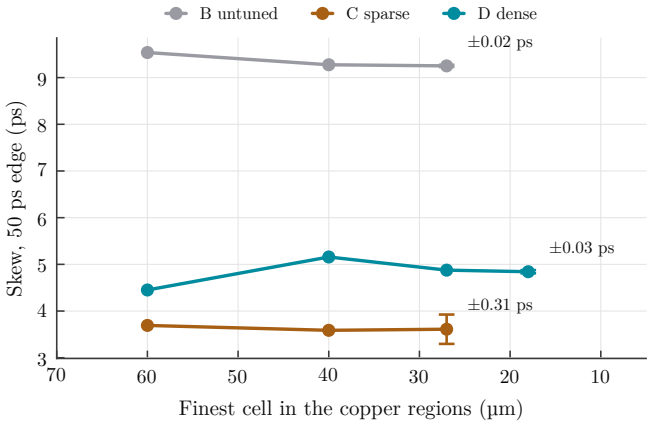


Figure 10: Skew against the finest cell size. The bar on each variant's finest point is its formal GCI bound (Section 6).

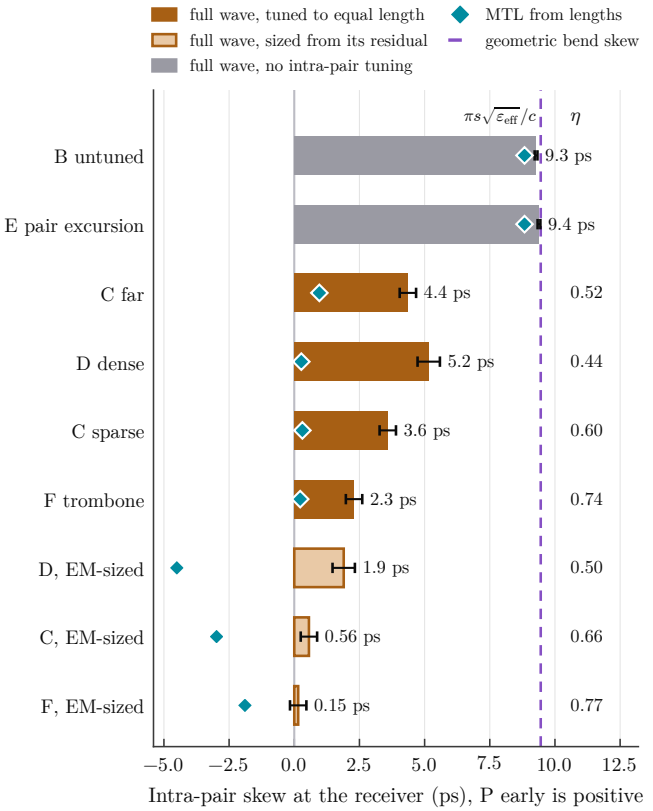


Figure 11: Skew at the receiver, 50 ps edge. Copper: tuned to exactly equal length. Grey: no intra-pair tuning. Diamonds: the transmission-line model with the same lengths. Bars: numerical uncertainty.

7.3 Where the delay goes

The gap is consistent with two effects that the line model leaves out. The serpentine experiment isolates them. It is a straight pair with 6 mm added to P in three excursions, once with folds 1.0 mm apart ($5h$) and once 0.15 mm apart, the clearance rule. The two have the same corners and the same added length. With loose

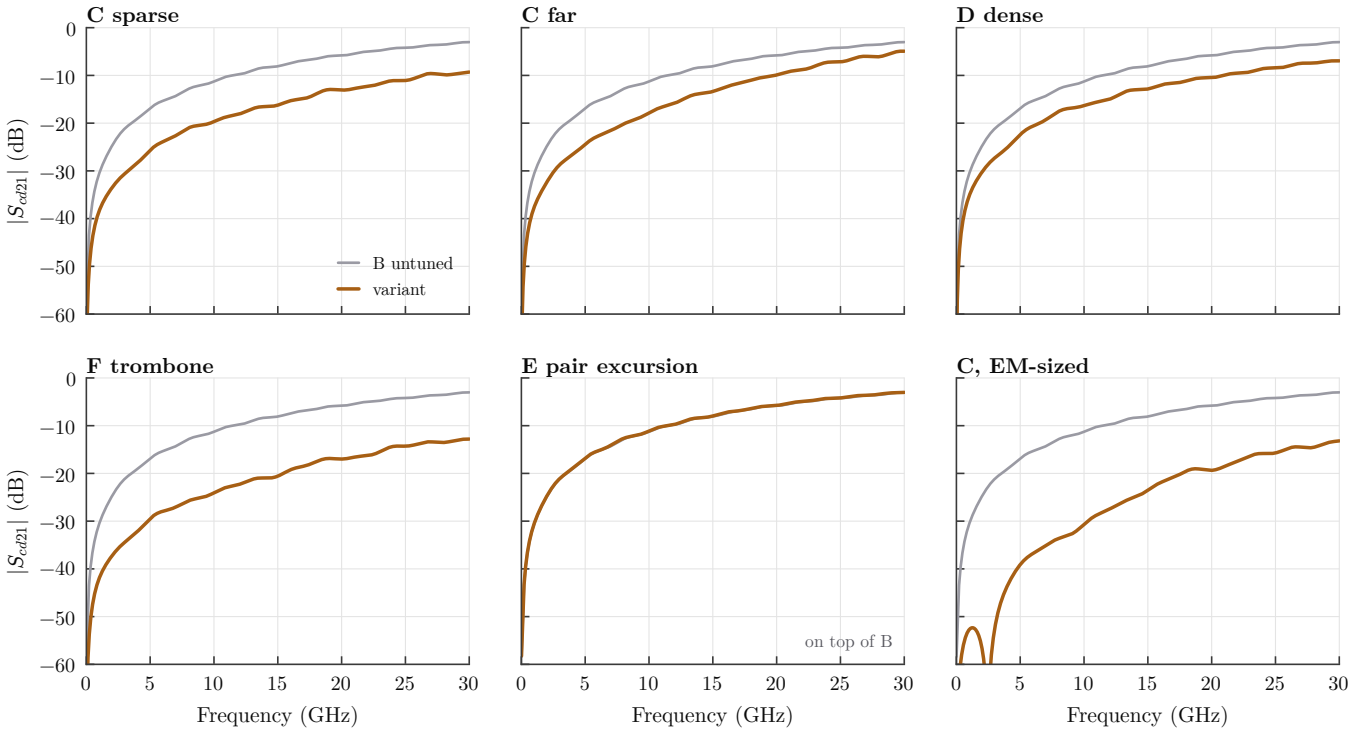


Figure 12: $|S_{cd21}|$ of each variant (copper) against the untuned bend B (grey).

folds $\eta = 0.84$ (0.84–0.91 across the meshes). With tight folds it is 0.68 (0.65–0.68), 16 to 24 points lower on every mesh (Fig. 13, Table 5). Coupling between the folds is the difference between them. The loose serpentine's own shortfall, which the line model also misses, we attribute to its twelve corners. In the bend corrections the shortfall per excursion, the skew the line model says the tuning removed less the skew full wave says it removed (net of the 0.44 ps the model misses on the bare bend), is 1.6, 1.4 and 1.1 ps for F, C and D. It is not a constant. It falls as the excursions get smaller and more numerous, but more slowly than their number grows, so the total shortfall still rises from one excursion to four. The three families change the excursion count, the amplitude and the fold spacing together, so this experiment does not separate them. The serpentines separate fold spacing alone, and the EM-sized variants of Section 8 test the amplitude. The dense accordion D has the most corners and the closest folds, and the lowest η.

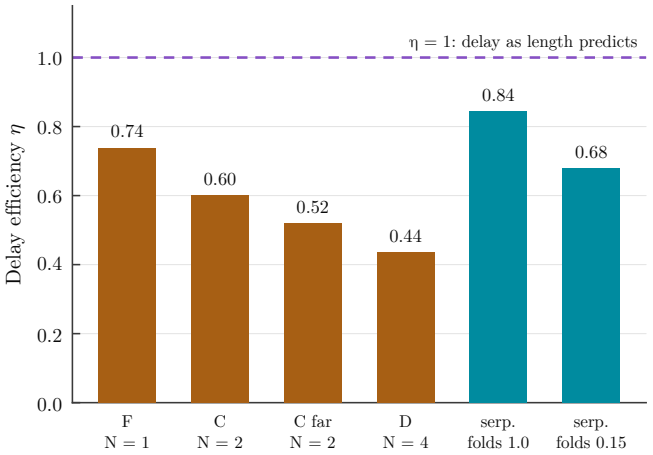


Figure 13: Delay efficiency η (Eq. 6) of the bend corrections (copper, N excursions) and of the 6 mm serpentines (teal) with loose and tight folds.

7.4 Stripline

The results above are all microstrip, where the odd and even modes travel at different velocities. To see whether that matters, we solved B, C, D, F and the serpentine set again in stripline, the same copper midway between two planes 0.7 mm apart with the ports to the lower plane. Stripline is homogeneous, so its nominal velocity is $c/\sqrt{\varepsilon_r}$ (6.917 ps/mm) and Eq. (6) uses that. Geometry predicts the bend to 0.6 % (10.97 ps against 11.04 ps). The length-matched tunings remove 81, 64 and 42 % of it for F, C and D, against 75, 61 and 44 % in microstrip, and the serpentines give $\eta = 0.84$ and 0.64, a fold penalty of 20 points against 16 in microstrip (Fig. 15). We had expected less, since in a homogeneous medium the far-end coupling between parallel lines vanishes. But the folds of a meander carry current in opposite directions, so what couples them acts backward along the route, and that coupling is present in both media. These stripline runs are on the medium mesh only.

7.5 Placement

C far is the same tuning as C, next to the launch instead of the bend. Its arrival skew is close (4.36 against 3.59 ps), but its mode conversion is not. Between the tuning and the bend the pair carries a skew in the other

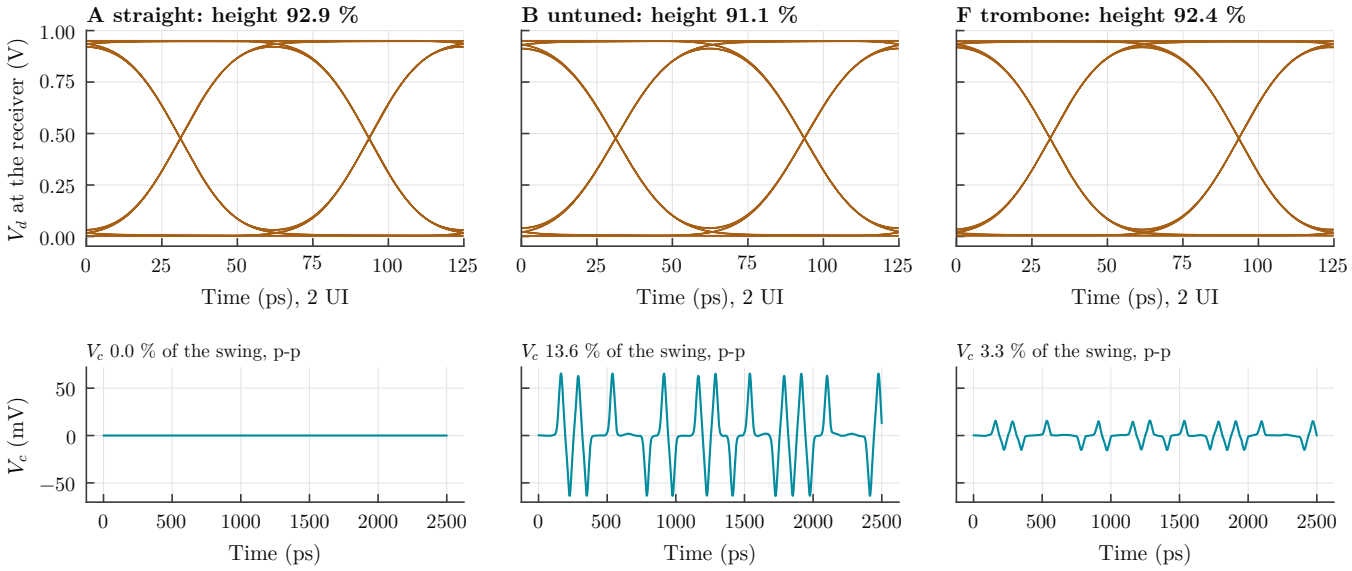


Figure 14: 16 Gb/s PRBS7 eyes (35 ps edges) and the common-mode voltage at the receiver.

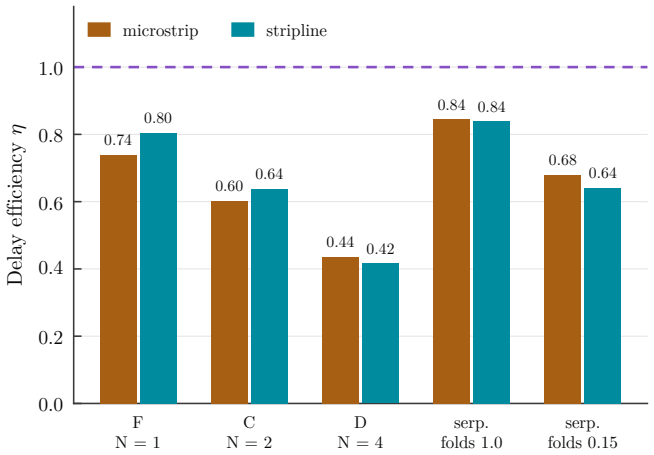


Figure 15: Delay efficiency of the same copper in microstrip and in stripline, each against its own nominal velocity.

direction, and the two conversions drift apart in phase with frequency. The frequency-resolved skew of C far climbs from 4.08 ps at 5 GHz to 6.49 ps at 25 GHz, while C stays near 3.55 ps (Fig. 16). Its peak $|S_{cd21}|$ to 20 GHz is -9.9 dB against -12.9 dB for C. Compensation belongs next to the mismatch.

7.6 Reflection, impedance and mode conversion

Where a trace leaves its partner, the differential impedance rises. The broad trombone F, whose P moves 0.886 mm further from N for 3 mm, lifts Z_{diff} to 114.4 Ω against 107.1 Ω for the untuned bend (Fig. 17), and has the worst return loss of the set (-16.8 dB against -20.7 dB, Fig. 18), so the variant with the least residual skew pays for it in reflection. The dense accordion barely moves Z_{diff}. Mode conversion follows the residual skew (Fig. 12), and E's curve lies on top of B's.

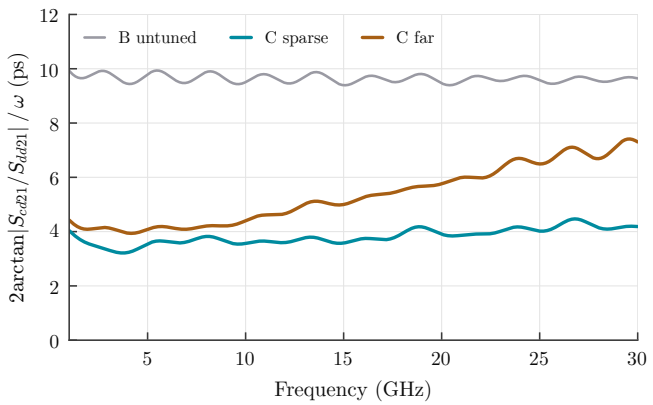


Figure 16: Frequency-resolved skew from Eq. (1). Compensation far from the bend (C far) holds at low frequency and fails as frequency rises.

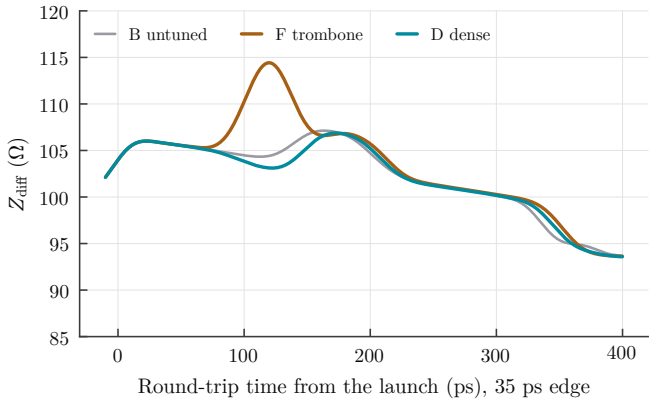


Figure 17: Differential TDR (35 ps edge). F's uncoupled stretch shows as the peak near 120 ps.

7.7 What the skew does to a signal

At 16 Gb/s with 35 ps edges, the untuned bend closes the differential eye from 92.9 % to 91.1 % of the swing and leaves its width at 0.936 UI (Fig. 14). The larger effect is common mode, 13.6 % of the swing peak to

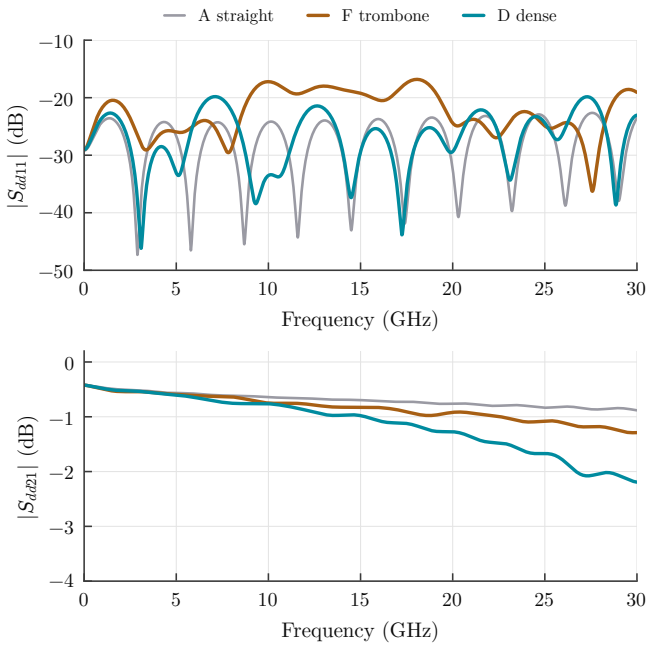


Figure 18: Differential return and insertion loss of the straight reference A, the trombone F and the dense accordion D.

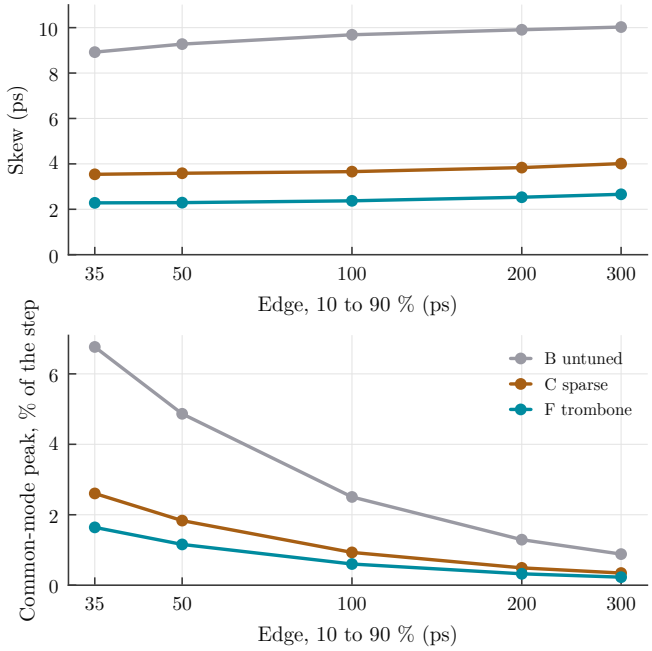


Figure 19: Skew and peak common-mode voltage against the edge rate.

peak at the receiver against 3.3 % for F. For a receiver with good common-mode rejection the skew of a single bend is a common-mode problem more than a timing one. Whether that common mode radiates depends on the structure beyond the pair, which this study does not model.

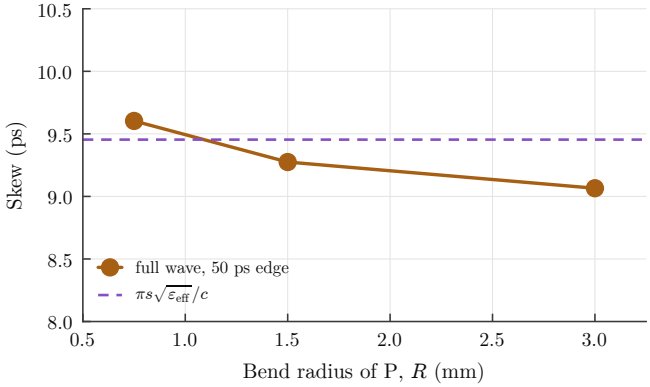


Figure 20: Skew of the untuned bend against its radius.

8 Optimization

EM-informed sizing. If each excursion loses a fixed amount of delay to its corners and folds, more amplitude on the same excursions should recover it. We lengthened C, D and F by their own medium-mesh residual, converted to length at the bend's rate ($\Delta L_{\text{add}} = \pi s\,(1 + t_{\text{skew}}/t_{\text{skew,B}})$), so that by CAD measure P is now longer than N, by 2.213, 2.484 and 1.991 mm in total added. By length alone these variants now overshoot, and the line model, which holds only lengths, predicts -1.89, -2.98 and -4.50 ps. Full wave gives 0.15 ps for F and 0.56 ps for C, within or close to their numerical bound, but 1.90 ps for D (sized from its medium-mesh residual, which Section 6 puts about 0.43 ps high, so if anything it was over-corrected). This is the test of the fixed-shortfall model, and it holds for the trombone, roughly for the sparse tuning, and not for the accordion. The extra millimetres delivered 92 % of their expected delay in F and 83 % in C, but only 62 % in D, whose lengthened verticals couple more strongly to one another. A broad or sparse tuning is calibrated by one field solution or one set of coupons per stackup, and a tuner could carry the per-excursion correction. A dense accordion needs iteration, which is one more reason to avoid it.

Pareto comparison. Over the bent candidates we minimise four objectives defined above, $|t_{\text{skew}}|$ (50 ps edge), RMS $|S_{cd21}|$, worst $|S_{dd11}|$ and tuning area, with no weighted score. The non-dominated set is {B, C, D, C (EM-sized), D (EM-sized), F (EM-sized)}. B is on it only because the untuned bend has no tuning area, and E, which adds area and removes nothing, is dominated by it. Fig. 21 shows two projections. Against area the dense accordion is cheap and weak, and against return loss the trombone trades reflection for skew.

9 Physical validation plan

This plan has not been executed. It calls for a coupon panel with one coupon per variant (A–F, the EM-sized variants and the serpentines), each in three copies, on one laminate and stackup, with identical launches, either edge-mount connectors or GSSG probe pads with ground

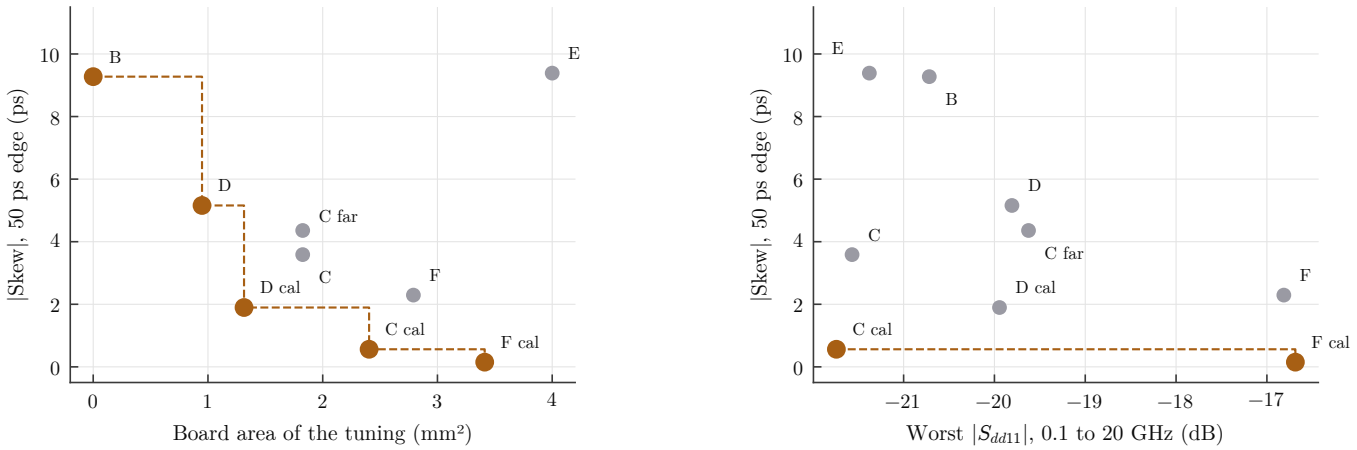


Figure 21: Two projections of the four-objective comparison. Copper points are non-dominated in that projection; the dashed line joins them.

vias, and a 2x-thru of the launch alone on the same panel for fixture de-embedding as in IEEE Std 370 [17]. Measure four-port S-parameters to 30 GHz on a calibrated VNA, de-embed to the launch plane, and microsection one coupon per panel to measure w, s, h and the copper. Extract ε_{eff} of the odd and even modes from coupon A and rerun the EM with the measured dimensions before comparing. The comparison is skew, η (Eq. 6, against B on the same panel), $|S_{dd11}|$, $|S_{cd21}|$ and TDR, with error bars from the spread of the three copies, the VNA's stated uncertainty and the GCI here. The predictions to test are that equal-length tuning leaves 2.29 to 5.16 ps at this pitch, that the shortfall grows with the excursion count, and that the same holds on a stripline panel. Near- or far-field scanning would be needed only if emission is the claim, which it is not here.

10 Recommendations

For edge-coupled pairs at this pitch and coupling, in microstrip and in stripline, and with the caveat that the numbers are simulated:

- Expect geometry to predict a bend's skew well (here within 1.9 % at a 50 ps edge, and 10.03 ps at 300 ps), and expect it to over-predict what a length-matched tuning removes. Budget a residual, or size the tuning from a field solution or a coupon.
- Prefer few excursions. The shortfall per excursion shrinks with its size, but not as fast as the count grows. A single excursion removed 75 % of the skew, four removed 44 %.
- Keep folds apart. Here folds at the clearance rule lost 16–24 points of efficiency against folds $5h$ apart.
- Tune next to the mismatch. The same correction at the launch matches arrival time but not phase at high frequency.
- Watch the impedance of a broad excursion. The uncoupled stretch lifts Z_{diff}, here to 114.4 Ω. Keep its amplitude small relative to the pitch or keep it short.
- A pair-level excursion does not fix intra-pair skew.
- None of this is special to microstrip. In stripline the bend's skew matched geometry to 0.6 %, the length-matched tunings removed a similar share, and tight folds cost 20 points. The edge rate and the interface decide whether a few picoseconds matter. At 16 Gb/s here they matter more as common mode than as eye closure.

11 Reproducibility

The study's repository, to be published with the report, holds `parameters.yaml`; `scripts/` with the geometry generator (and its KiCad export and DRC), the openEMS model writer, the run orchestrator with provenance, the S-parameter extraction, the line model, the analysis, the convergence study and the figure and table generators; and `figures/`. Every number in this report is a macro generated from `build/results/results.json`. We made the 140 runs on one Linux host, 27.7 h of wall time in all on single-threaded solver processes. The `README.md` lists the commands, and regenerating every figure and table from the stored S-parameters takes a few minutes.

12 Limitations

One width and one pitch, in one microstrip and one stripline stackup. The stripline ports run to the lower plane only, and its plane spacing came from a closed-form estimate. Its Z_{diff} is about 95.6 Ω by TDR, so its absolute return loss is worse than microstrip's, which does not affect the skew comparisons. The return plane is a perfect conductor and the copper a conducting sheet, so conductor loss is approximate and roughness is absent. The dielectric loss model has $\tan\delta \propto 1/f$. Lumped ports at the

launch are not a connector or a probe. Their reflection is common to every variant and cancels in the comparisons, but not in absolute return loss. The staircased mesh resolves the smallest features (the 75 μm chamfers) with under two cells (1.9) at the medium level, and the convergence study bounds the effect for B, C, D and the serpentines only, and the stripline set was solved on one mesh. Edges faster than 35 ps and frequencies above 30 GHz are outside the simulated band. Common-mode voltage and S_{cd21} are not emissions.

13 Conclusions

At a 20 mil pitch a U-bend's skew is what geometry says, about 9 ps here. Matching the lengths does not remove it. A length-matched serpentine removed between 44 and 75 % of the skew in full-wave simulation, and the remainder grew with the number of excursions and the closeness of the folds. A line model with the same lengths missed nearly all of it, and the same copper in stripline behaved the same way. One field-solver iteration sized the broad and sparse tunings to within 0.15 and 0.56 ps, and the dense accordion needed more. More length matching will not close the gap. Size the tuning by what it does electrically, and keep it next to the bend with few excursions set well apart.

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